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Hybrid riser for archery bow

Abstract

In some embodiments, an archery bow riser comprises a body defining a first end, a second end and a grip portion. The body defines a shooting axis and comprises a first portion and a second portion. The first portion comprises a metal and the second portion comprises carbon fibers. The first portion extends to a first side of the shooting axis and to a second side of the shooting axis, and the second portion extends to the first side of the shooting axis and to the second side of the shooting axis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. patent application Ser. No. 17/883,463, filed Aug. 8, 2022, which claims the benefit of U.S. patent application Ser. No. 16/950,706, filed Nov. 17, 2020, which claims the benefit of U.S. patent application Ser. No. 16/352,719, filed Mar. 13, 2019, which claims the benefit of U.S. Patent Application No. 62/642,468, filed Mar. 13, 2018, the entire contents of each of which are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) This invention relates generally to archery bows and more specifically to riser and frame constructions for archery bows. Archery bows generally include a riser, which forms the main structure of the bow. Known risers are often made from a metal such as aluminum, or alternatively

from a composite material such as a polymer reinforced with glass or carbon fibers. Metal risers have certain benefits and drawbacks. Metal risers can be cut to any shape but can tend to be more flexible than composite risers. Screw threads can be formed directly in a metal, and fasteners used in the bow can engage the metal directly. Composite risers have different benefits and different drawbacks. Composite risers are typically formed in a mold, which can generally produce risers having only one shape; however, composite risers can have superior strength and deflection characteristics.

(2) There remains a need for novel bow and riser designs that improve the bow by providing increased stiffness and/or lighter weight than prior designs.

(3) All US patents and applications and all other published documents mentioned anywhere in this application are incorporated herein by reference in their entirety.

(4) Without limiting the scope of the invention a brief summary of some of the claimed embodiments of the invention is set forth below. Additional details of the summarized embodiments of the invention and/or additional embodiments of the invention may be found in the Detailed Description of the Invention below.

(5) A brief abstract of the technical disclosure in the specification is provided as well only for the purposes of complying with 37 C.F.R. 1.72. The abstract is not intended to be used for interpreting the scope of the claims.

SUMMARY OF THE INVENTION

(6) In some embodiments, an archery bow riser comprises a body defining a first end, a second end and a grip portion. The body defines a shooting axis and comprises a first portion and a second portion. The first portion comprises a metal and the second portion comprises carbon fibers. The first portion extends to a first side of the shooting axis and to a second side of the shooting axis, and the second portion extends to the first side of the shooting axis and to the second side of the shooting axis.

(7) In some embodiments, second portion extends continuously for at least one-quarter of a length of the riser. In some embodiments, second portion extends continuously for at least one-half of a length of the riser.

(8) In some embodiments, the body comprises a first limb engagement location and a second limb engagement location. The first portion extends continuously from the first limb engagement location to the second limb engagement location. The second portion extends continuously from the first limb engagement location to the second limb engagement location.

(9) In some embodiments, an archery bow comprises a riser engaged with a first limb fastener and a second limb fastener. The riser comprises a first portion comprising metal and a second portion comprising carbon fiber. The riser comprises a tension surface and a compression surface. The second portion receives tensile forces from the first limb fastener and from the second limb fastener. The compression surface excludes the second portion.

(10) In some embodiments, the second portion comprises a first aperture arranged to receive the first limb fastener and a second aperture arranged to receive the second limb fastener. In some embodiments, the first portion comprises a limb nut aperture aligned with the first aperture.

(11) In some embodiments, an archery bow comprises a riser, a first limb supported by the riser at a first support location and a second limb supported by the riser at a second support location. The riser comprises a first portion comprising a first material and a second portion comprising a second material, the second material having a higher elastic modulus than the first material. The first portion extends from the first support location to the second support location. The first portion is arranged to receive compressive forces from the first and second limbs. The second portion extends from the first support location to the second support location. The second portion is arranged to receive tensile forces from the first and second limbs.

(12) In some embodiments, the first limb is attached to the riser by a limb bolt and a limb nut, and the first portion comprises a first limb nut aperture. In some embodiments, the second portion

comprises a second limb nut aperture aligned with the first limb nut aperture. In some embodiments, the second portion comprises a limb bolt aperture. In some embodiments, the first portion comprises a third limb nut aperture aligned with the first limb nut aperture. In some embodiments, the first portion comprises a slot and the second portion is oriented in the slot.

(13) These and other embodiments which characterize the invention are pointed out with particularity in the claims annexed hereto and forming a part hereof. However, for a better understanding of the invention, its advantages and objectives obtained by its use, reference can be made to the drawings which form a further part hereof and the accompanying descriptive matter, in which there are illustrated and described various embodiments of the invention.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) A detailed description of the invention is hereafter described with specific reference being made to the drawings.
- (2) FIGS. **1-3** show different views of an embodiment of an archery bow.
- (3) FIGS. **4** and **5** show an exploded view of an archery bow riser.
- (4) FIGS. **6-8** show views of another embodiment of an archery bow riser.
- (5) FIGS. **9-12** show exploded views of the riser shown in FIGS. **6-8**.
- (6) FIGS. **13-15** show the riser of FIGS. **6-8** engaged with limb fasteners.
- (7) FIGS. **16-19** show another embodiment of a riser.
- (8) FIGS. **20-23** show another embodiment of a riser.

DETAILED DESCRIPTION OF THE INVENTION

- (9) While this invention may be embodied in many different forms, there are described in detail herein specific embodiments of the invention. This description is an exemplification of the principles of the invention and is not intended to limit the invention to the particular embodiments illustrated.
- (10) For the purposes of this disclosure, like reference numerals in the figures shall refer to like features unless otherwise indicated.
- (11) FIGS. **1-3** show an embodiment of an archery bow **10**. In some embodiments, a bow **10** comprises a riser **12**, a first limb **14** and a second limb **16**. The first limb **14** supports a first rotatable member **20** and the second limb **16** supports a second rotatable member **30**. In some embodiments, a limb (e.g. first limb **14**) comprises a limb assembly having multiple limb members (e.g. **14a**, **14b**) as shown in FIG. **1**.
- (12) In some embodiments, the limbs **14**, **16** are attached to the riser **12** by limb cups **22**, **24**. In some embodiments, limbs **14**, **16** and limb cups **22**, **24** comprise features as disclosed in U.S. Pat. No. 8,453,635.
- (13) In some embodiments, the forces transferred between a limb **14** and the riser **12** comprise a moment connection, for example comprising a force couple comprising a compressive force C and a tensile force T. In some embodiments, a compressive force C is applied to the riser **12** by a limb cup **22**. In some embodiments, a tensile force T is applied via a limb fastener **24** such as a limb bolt **35**. In some embodiments, a head of a limb bolt **35** engages a limb cup **22**. In some embodiments, a limb bolt **35** is attached to the riser **12**, for example engaging a threaded hole in the riser **12**. In some embodiments, the limb bolt **35** engages a limb nut **36** that comprises a threaded hole. In some embodiments, a limb nut **36** comprises a circular cross-sectional shape or a cylindrical outer shape, and is received in a cavity in the riser **12**, which allows limb nut **36** to rotate and the angle of the limb bolt **35** with respect to the riser **12** to change, for example as the limb bolt **35** is adjusted.
- (14) Desirably, the bow **10** defines a shooting axis **72**. In some embodiments, the bow **10** comprises a grip **11**. In some embodiments, the grip **11** is attached to the riser **12** or comprises a portion of the

riser **12**. When shooting the bow **10**, a shooter will typically grasp the grip **11** and apply a force F to the grip **11**/riser **12**. In some embodiments, the riser **12** acts as a beam and carries stresses, which may originate both from the bow's **10** internal forces and any from external force F applied by a shooter. Forces present in a riser **12** can primarily comprise tensile forces on the front side **76** of the riser **12** and compressive forces on the back side **77** of the riser **12**. In some embodiments, a front surface of the riser **12** comprises a major tension surface **66** and a back surface **77** of the riser comprises a major compression surface.

(15) In some embodiments, the bow **10** comprises a cable guard **28** arranged to move cables laterally, for example to position the cables away from the shooting axis **76**. In some embodiments, the cable guard **28** is supported by the riser **12** and places a torqueing force on the riser **12**. In some embodiments, forces from the cable guard **28** place one side of the riser **12** in tension and one side of the riser **12** in compression. In some embodiments, the riser **12** comprises a lateral tension surface **68**.

(16) In some embodiments, the riser **12** comprises a first portion **40** comprising a first material and a second portion **50** comprising a second material. In some embodiments, the second portion **50** comprises a material having a higher elastic modulus than the material of the first portion **40**. In some embodiments, the second portion **50** has a greater stiffness than the first portion **40**. In some embodiments, the first portion **40** consists of the first material and comprises a single piece of material. In some embodiments, the second portion **50** consists of the second material and comprises a single piece of material.

(17) In some embodiments, the first portion **40** is formed from metal. Any suitable metal can be used. In some embodiments, the first portion **40** comprises aluminum.

(18) In some embodiments, the second portion **50** comprises reinforcing fibers in a matrix material. The reinforcing fibers can comprise any suitable type of reinforcing material, such as carbon fiber, glass fiber, aramid fiber or the like and suitable combinations thereof. In some embodiments, the second portion **50** comprises carbon fiber in a resin matrix. In some embodiments, the reinforcing fibers of the second portion **50** consist of carbon fiber.

(19) In some embodiments, the second portion **50** has a greater resistance to tensile deformation than the first portion **40**, and the second portion **50** comprises a tensile stiffener that reinforces the riser **12** against deformation. In some embodiments, the second portion **50** is referred to herein as a stiffener **50**.

(20) FIGS. **4** and **5** show views of the riser **12** shown in FIGS. **1-3** with the second portion **50** removed from the first portion **40**.

(21) In some embodiments, the first portion **40** of the riser **12** comprises a cavity **74**. In some embodiments, the stiffener **50** is oriented in the cavity **74**. In some embodiments, a size and shape of the cavity **74** is similar to the size and shape of the stiffener **50**. In some embodiments, a stiffener **50** extends to the front surface **76** of the riser **12**, or the front surface **76** of the first portion **40**. In some embodiments, a front surface **56** of the stiffener **50** is aligned with the front surface **76** of the first portion **40**. In some embodiments, the stiffener **50** extends to the major tension surface **66** of the riser **12**. In some embodiments, the major tension surface **66** comprises the stiffener **50**.

(22) In some embodiments, the major tension surface **66** comprises both the first portion **40** and the second portion **50**. In some embodiments, the major compression surface **77** comprises the first portion **40** and excludes the second portion **50**.

(23) The second portion **50** can be attached to the first portion **40** using any suitable method. In some embodiments, an adhesive is used. In some embodiments, multiple surfaces of the second portion **50** are attached to multiple surfaces of the first portion **40** with an adhesive. Any suitable adhesive can be used that will attach material of the second portion **50** to material of the first portion **40**. In some embodiments, an adhesive comprises an epoxy, an epoxy mixture, an acrylic, a methacrylate and/or any other suitable adhesive.

(24) In some embodiments, one or more fasteners can be used to attach the second portion **50** to the

first portion **40**. In some embodiments, the second portion **50** can comprise an aperture. In some embodiments, a fastener with a head extends through an aperture of the second portion **50** and engages the first portion **40**. In some embodiments, a pin can engage (e.g. frictionally) the first portion **40** and the second portion **50**. Any suitable combination of attachment methods can be used. In some embodiments, the first and second portions **40**, **50** are attached to one another by at least one fastener and by an adhesive.

(25) Any suitable number of surfaces of the second portion **50** can be attached to the first portion **40** by adhesive. In some embodiments, the second portion **50** comprises six sides, and any suitable number of sides can be attached by adhesive. As shown in FIGS. **4** and **5**, all sides of the second portion **50** except the front surface **56** can be attached to the first portion **40**.

(26) The first portion **40** can be made using any suitable method. In some embodiments, the first portion **40** is cast in a mold. In some embodiments, the first portion **40** is cut from a workpiece.

(27) The second portion **50** can be made using any suitable method. In some embodiments, the second portion **50** is formed in a mold. In some embodiments, the second portion **50** is formed in the cavity **74** using the cavity **74** as a mold. In some embodiments, an uncured slurry of matrix material and reinforcing fibers is deposited into the cavity **74** and then cured. In some embodiments, interior walls of the cavity **74** comprise a portion of a mold, and another exterior mold portion can be used to close the cavity **74** and allow injection molding under heat and/or pressure.

(28) In some embodiments, a stiffener **50** comprises a first length portion **51** and a second length portion **52**. In some embodiments, a longitudinal axis of the first length portion **51** is parallel to and offset from a longitudinal axis of the second portion **52**. In some embodiments, the stiffener **50** comprises one or more curved portions **53**.

(29) In some embodiments, the riser **12** comprises an offset portion **15** that defines a sight window **13**. Desirably, the offset portion **15** is offset from the shooting axis **76**. In some embodiments, the shooting axis **76** extends through the sight window **13**. In some embodiments, the first portion **40** of the riser **12** extends above and below the shooting axis **76**. In some embodiments, the stiffener **50** extends above and below the shooting axis **76**. In some embodiments, the riser **12** first portion **40** and the stiffener **50** are attached to one another continuously and extend above and below the shooting axis **76**.

(30) In some embodiments, the stiffener **50** occupies at least a quarter of a length dimension (e.g. height dimension) of the riser **12**. In some embodiments, the stiffener **50** occupies approximately one-third of the length dimension of the riser **12**. In some embodiments, the stiffener **50** occupies approximately one-half of the length dimension of the riser **12**. In some embodiments, the stiffener **50** does not extend to the limb connections.

(31) In some embodiments, the riser **12** comprises a grip **18**. In some embodiments, the grip **18** is positioned adjacent to a portion of the stiffener **50**. In some embodiments, the grip **18** defines a cavity, and the riser **12** first portion **40** and the stiffener **50** are oriented in the cavity.

(32) In some embodiments, the first portion **40** comprises portions positioned around the shooting axis **72**. In some embodiments, the first portion **40** comprises portions located to a first side and to a second side (e.g. opposite sides) of the shooting axis **72**. In some embodiments, the first portion **40** comprises portions located above and below the shooting axis **72**.

(33) In some embodiments, the second portion **50** comprises portions positioned around the shooting axis **72**. In some embodiments, the first portion **40** comprises portions located to a first side and to a second side (e.g. opposite sides) of the shooting axis **72**. In some embodiments, the first portion **40** comprises portions located above and below the shooting axis **72**.

(34) The second portion **50** can extend for any suitable length portion of the riser **12**. In some embodiments, the second portion **50** extends continuously for one-quarter of a length of the riser **12**. In some embodiments, the second portion **50** extends continuously for one-third of a length of the riser **12**. In some embodiments, the second portion **50** extends continuously for one-half of a

length of the riser **12**.

(35) FIGS. **6-12** show views of another embodiment of a riser **12**. In some embodiments, the riser **12** comprises a first portion **40** and a second portion **50**. In some embodiments, the second portion **50** occupies a majority of a length dimension of the riser **12**.

(36) In some embodiments, the second portion **50** extends continuously between the limb **14**, **16** connections. In some embodiments, the second portion **50** engages limb fasteners (e.g. **24**).

(37) In some embodiments, the second portion **50** comprises a portion of the major tension surface **66** of the riser **12**. In some embodiments, the second portion comprises a portion of the lateral tension surface **68**. In some embodiments, the second portion **50** comprises a single piece of material that comprises the major tension surface **66** and the lateral tension surface **68** of the riser **12**. In some embodiments, the second portion **50** occupies a front-lateral tension quadrant of the riser **12** cross-section.

(38) In some embodiments, all contacting surfaces between the first portion **40** and the second portion **50** of the riser **12** can be attached using an adhesive. In some embodiments, the first portion **40** and the second portion **50** are attached via pins, fasteners, etc. In some embodiments, a pin can extend through apertures in the first portion **40** and the second portion **50**, for example frictionally engaging the apertures. As shown in FIGS. **6-12**, the second portion **50** is arranged to receive fasteners. In some embodiments, a fastener can abut the second portion **50** and engage the first portion **40**. In some embodiments, the first portion **40** comprises threaded apertures **42** arranged to engage a fastener. In some embodiments, the second portion **50** comprises apertures **58** arranged to receive a fastener. In some embodiments, apertures **58** in the second portion **50** are countersunk and provide a cavity for a head of a fastener.

(39) In some embodiments, the cavity **74** defined by the first portion **40** comprises a first slot portion **44**. In some embodiments, the cavity **74** comprises a second slot portion **45**. In some embodiments, the first and second slot portions **44**, **45** are located at or near opposite ends of the riser **12**. In some embodiments, a slot portion **44**, **45** is defined by a first sidewall **46** and a second sidewall **47**.

(40) In some embodiments, the second portion **50** comprises a first tab portion **54** and a second tab portion **55**. In some embodiments, the first tab portion **54** of the second portion **50** is oriented in the first slot portion **44** of the first portion **40**. In some embodiments, the second tab portion **55** of the second portion **50** is oriented in the second slot portion **45** of the first portion **40**.

(41) In some embodiments, surfaces of the first portion **40** and second portion **50** that contact one another are shaped to comprise complimentary mating surfaces. In some embodiments, such contacting surfaces of the first portion **40** and second portion **50** are attached with an adhesive. In some embodiments, the contacting surfaces of the first portion **40** and second portion **50** comprise mating protrusion and recess structures to aid in both alignment and force transfer between the portions **40**, **50**. As shown in the FIG. **12**, in some embodiments, the first portion **40** comprises a protrusion **70** having a predetermined shape, and the second portion **50** comprises a mating recess **71**.

(42) FIGS. **13-15** show an embodiment of a riser **12** and associated limb fasteners **34**, such as a limb bolt **35** and a limb nut **36**. FIG. **15** shows a limb cup **22** attached to the riser **12** via the limb fasteners **34**.

(43) Referring to FIGS. **9-15**, desirably, the riser **12** is arranged to engage limb fasteners **34**. In some embodiments, the first portion **40** engages a limb fastener **34**. In some embodiments, the second portion **50** engages a limb fastener **34**. In some embodiments, both the first portion **40** and the second portion **50** engage a limb fastener **34**.

(44) In some embodiments, the second portion **50** comprises one or more limb fastener apertures **60**, which are constructed and arranged to engage a limb fastener **34**. In some embodiments, the second portion **50** comprises a limb nut aperture **62** that is arranged to receive a limb nut **36**. In some embodiments, the second portion **50** comprises a limb bolt aperture **61** that is arranged to

receive a limb bolt **35**. In some embodiments, the first tab portion **54** of the second portion **50** comprises limb fastener apertures **60**. In some embodiments, the second tab portion **55** comprises a second set of limb fastener apertures **60**.

(45) In some embodiments, tensile forces applied by the first limb **14** are transferred to the second portion **50** by limb fasteners **34**. In some embodiments, forces from the first limb **14** are applied to a first end of the second portion **50**, for example being applied to the first tab portion **54**. In some embodiments, tensile forces applied by the second limb **16** are transferred to the second portion **50** by a second set of limb fasteners **34**. In some embodiments, forces from the second limb **16** are applied to a second end of the second portion **50**, for example being applied to the second tab portion **55**. In some embodiments, the second portion **50** comprises a continuous member that receives tensile forces applied by both limbs **14**, **16**. In some embodiments, the second portion comprises a single piece of material.

(46) In some embodiments, the first portion **40** is arranged to engage a limb fastener **34**. In some embodiments, the first portion **40** comprises a limb nut aperture **64** arranged to receive a limb nut **36**. In some embodiments, a limb nut aperture **64** is located near an end of the first portion **40**. In some embodiments, the limb nut aperture **64** is located in the first sidewall **46** that defines the first slot portion **44**. In some embodiments, the first portion **40** comprises a second limb nut aperture **65** in proximity to the first limb nut aperture **64**. In some embodiments, the second limb nut aperture **65** is located in the second sidewall **47** that defines the first slot portion **44**. In some embodiments, the limb nut apertures **64**, **65** near the first slot portion **44** are aligned with one another. In some embodiments, a second end of the first portion **40** is configured similarly to the first end, for example providing aligned limb nut apertures **64**, **65** near the second slot portion **45**.

(47) In some embodiments, the tabs **54**, **55** of the second portion **50** are oriented in the slot portions **44**, **45** of the first portion **40**. In some embodiments, one or more limb nut apertures **64**, **65** of the first portion are aligned with a limb nut aperture **62** of the second portion. For example, the limb nut apertures **62**, **64**, **65** can comprise a similar shape and be aligned on a common axis. Thus, in some embodiments, both the first portion **40** and the second portion **50** engage a limb fastener **34**.

(48) In some embodiments, a limb nut **36** comprises a circular cross-sectional shape. In some embodiments, a limb nut **36** comprises a cylindrical outer shape. In some embodiments, limb nut apertures (e.g. **62**, **64**, **65**) comprise a circular shape or circular shaped portions that receive the limb nut **36** and allow the limb nut **36** to rotate within the limb nut aperture **62**, **64**, **65**.

(49) In some embodiments, the second portion **50** comprises a material having a higher elastic modulus than the material of the first portion **40**.

(50) The second portion **50** as described herein is well suited to comprise a tension reinforcing member. In some embodiments, the second portion comprises carbon fiber, and is used in conjunction with a first portion **40** that is not carbon fiber. In some embodiments, the first portion **40** comprises a metal. Any suitable metal can be used, and relatively lightweight metals such as aluminum may be preferred.

(51) In some embodiments, the second portion **50** comprises carbon fibers in a resin matrix. Any suitable construction comprising carbon fibers can be used. Carbon fibers can be produced using any suitable method including the use of polyacrylonitrile (PAN), the processing of pitch or any other suitable method. The carbon fibers and any suitable filler material, such as a resin, can be used to form a second portion **50** having any suitable size, shape and configuration. In some embodiments, the materials are cured in a mold. In some embodiments, the second portion **50** comprises layers of carbon fiber, and the fibers of adjacent layers are oriented in different directions. In some embodiments, the second portion **50** comprises carbon fibers in a weave pattern.

(52) In some embodiments, the first portion **40** comprises a single piece of material. In some embodiments, the second portion **50** comprises a single piece of material.

(53) FIGS. **16-19** show views of another embodiment of a riser **12**. In some embodiments, a riser

12 comprises a tension rail **80** that extends continuously along a tension face **85** of the riser **12**. In some embodiments, the tension rail **80** extends continuously from a first location near one limb nut aperture **64** to a second location near another limb nut aperture **64**.

(54) In some embodiments, the tension rail **80** comprises a first portion **81** that consists of the first portion **40** of the riser **12**.

(55) In some embodiments, the tension rail **80** comprises a second portion **82** that consists of the second portion **50** of the riser **12**.

(56) In some embodiments, the tension rail **80** comprises a third portion **83** that comprises the first portion **40** and the second portion **50** of the riser **12**.

(57) In some embodiments, the tension rail **80** comprises a fourth portion **84** that consists of the first portion **40** of the riser **12**.

(58) In some embodiments, the first portion **40** comprises a plurality of weight reducing apertures **90**. The weight reducing apertures **90** can have any suitable size and shape. The weight reducing apertures **90** are typically included in interior portions of the riser **12** and do not extend into the tension rail **80** or a compression rail **88** located on a compression side of the riser **12**.

(59) In some embodiments, the second portion **50** comprises one or more weight reducing apertures **92**. The weight reducing apertures **92** can have any suitable size and shape.

(60) In some embodiments, weight reducing apertures **92** of the second portion **50** have sizes and shapes similar to weight reducing apertures **90** of the first portion **40**. In some embodiments, similarly shaped weight reducing apertures **90**, **92** are located adjacent one another and create a riser **12** that comprises a first portion **40** and a second portion **50**, but has an appearance similar to a traditional riser.

(61) FIGS. **20-23** show another embodiment of a riser **12** comprising a first portion **40** and a second portion **50**. In some embodiments, the first portion **40** comprises a first material, such as metal, and the second portion **50** comprises a second material, such as a non-metal.

(62) In some embodiments, a bow defines a bowstring plane **73**, wherein a bowstring of the bow theoretically travels in the bowstring plane **73** as the bow is drawn and fired. Desirably, the shooting axis **72** of the bow is positioned in the bowstring plane **73**.

(63) In some embodiments, the riser **12** defines a sight window **13**. In some embodiments, the sight window **13** is aligned upon the bowstring plane **73** and the riser **12** comprises structure positioned lateral to the sight window **12**. In some embodiments, the first portion **40** comprises the structure extending adjacent to the sight window **13**.

(64) In some embodiments, the second portion **50** is positioned below the sight window **13**. In some embodiments, a portion of the second portion **50** overlaps the bowstring plane **73**.

(65) In some embodiments, the riser **12** comprises a grip area **19** comprising structure defining an aperture **86**. In some embodiments, a portion of a shooter's hand is oriented in the aperture **86** when the riser **12** is grasped. In some embodiments, the aperture **86** is partially defined by the first portion **40** of the riser **12** and is partially defined by the second portion **50** of the riser **12**. In some embodiments, the first portion **40** is located to a first side of the aperture **86** and the second portion **50** is located to a second side of the aperture **86**. In some embodiments, the first portion **40** is located to a first side of the bowstring plane **73** and the second portion **50** is located to a second side of the bowstring plane **73**.

(66) The risers **12** disclosed herein can be used in any type of bow, including all types of compound bows (e.g. single cam, dual cam, etc.) and non-compound bows (e.g. traditional and recurve).

(67) The above disclosure is intended to be illustrative and not exhaustive. This description will suggest many variations and alternatives to one of ordinary skill in this field of art. All these alternatives and variations are intended to be included within the scope of the claims where the term "comprising" means "including, but not limited to." Those familiar with the art may recognize other equivalents to the specific embodiments described herein which equivalents are also intended to be encompassed by the claims.

(68) Further, the particular features presented in the dependent claims can be combined with each other in other manners within the scope of the invention such that the invention should be recognized as also specifically directed to other embodiments having any other possible combination of the features of the dependent claims. For instance, for purposes of claim publication, any dependent claim which follows should be taken as alternatively written in a multiple dependent form from all prior claims which possess all antecedents referenced in such dependent claim if such multiple dependent format is an accepted format within the jurisdiction (e.g. each claim depending directly from claim 1 should be alternatively taken as depending from all previous claims). In jurisdictions where multiple dependent claim formats are restricted, the following dependent claims should each be also taken as alternatively written in each singly dependent claim format which creates a dependency from a prior antecedent-possessing claim other than the specific claim listed in such dependent claim below.

(69) This completes the description of the preferred and alternate embodiments of the invention. Those skilled in the art may recognize other equivalents to the specific embodiment described herein which equivalents are intended to be encompassed by the claims attached hereto.

Claims

1. An archery bow comprising: a riser engaged with a first limb fastener and a second limb fastener, the riser comprising metal and carbon fiber, the first limb fastener engaged with the metal, the second limb fastener engaged with the metal, the riser comprising a major tension surface and a major compression surface, the major tension surface comprising the metal, the major tension surface comprising the carbon fiber.
2. The archery bow of claim 1, the major compression surface comprising the metal.
3. The archery bow of claim 2, the major compression surface comprising the carbon fiber.
4. The archery bow of claim 1, the metal comprising a first aperture arranged to receive the first limb fastener and a second aperture arranged to receive the second limb fastener.
5. The archery bow of claim 4, wherein the first limb fastener comprises a first limb nut, the bow further comprising a limb bolt attached to the first limb nut.
6. The archery bow of claim 4, the metal comprising a limb nut aperture aligned with the first aperture.
7. The archery bow of claim 1, comprising limbs arranged to apply tension and compression to the riser, the metal arranged to receive the tension and the compression.
8. The archery bow of claim 7, the carbon fiber arranged to receive the tension.
9. The archery bow of claim 8, the carbon fiber arranged to receive the compression.
10. The archery bow of claim 1, comprising an adhesive attaching the metal to the carbon fiber.
11. The archery bow of claim 1, the carbon fiber comprising a first portion and a second portion, the first portion positioned to a first side of a shooting axis, the second portion positioned to a second side of the shooting axis.
12. An archery bow comprising: a riser; a first limb supported by the riser at a first support location and engaged with the riser by a first limb fastener; a second limb supported by the riser at a second support location and engaged with the riser by a second limb fastener, the riser comprising metal and carbon fiber, the metal engaged with the first limb fastener, the first limb and the second limb arranged to apply tensile forces and compressive forces to the riser; the metal arranged to receive the tensile forces, the metal arranged to receive the compressive forces; the carbon fiber arranged to receive the tensile forces.
13. The archery bow of claim 12, the first limb fastener comprising a limb bolt and a limb nut, the metal comprising a first limb nut aperture.
14. The archery bow of claim 13, the second limb fastener comprising a second limb bolt and a second limb nut, the metal comprising a second limb nut aperture.

15. The archery bow of claim 14, the metal comprising a limb bolt aperture.
 16. The archery bow of claim 12, comprising an adhesive attaching the metal to the carbon fiber.
 17. The archery bow of claim 12, the riser comprising a major tension surface, the major tension surface comprising the metal.
 18. The archery bow of claim 17, the major tension surface comprising the carbon fiber.
 19. The archery bow of claim 18, the riser comprising a major compression surface, the major compression surface comprising the carbon fiber.
 20. The archery bow of claim 12, the carbon fiber comprising a first portion and a second portion, the first portion positioned to a first side of a shooting axis, the second portion positioned to a second side of the shooting axis.
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